Sentiment Analysis Based on Bangla Video Gaming Reviews Using Machine Learning and Explainable NLP

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***Abstract*—This research represents an innovative approach to sentiment and emotion analysis of Bangla comments from youtube gaming channels, addressing the challenges posed by linguistic nuances and the absence of standardized resources. Leveraging a dataset comprising 45,722 Bangla comments from diverse gaming channels, annotated for sentiment and emotion, we propose a deep learning-based stacking model achieving 96% accuracy. Our methodology includes data collection, compilation, preprocessing, and classification using multinomial naive bayes, recurrent convolutional neural network, and feedforward neural network models. Results analysis, including explainable ai techniques, highlights model performance and areas for improvement. Future work aims to enrich datasets, explore hybrid techniques, and implement transformer-based learning.**

***Index Terms*—Bangla sentiment analysis, emotion analysis, YouTube gaming channels, deep learning, stacking model, explainable ai, data preprocessing, classification model.**

I. Introduction

Sentiment and emotion analysis, integral to natural language processing [1], aims to extract subjective states from textual content. While extensively studied for languages like English, research in this domain for Bangla, one of the most widely spoken languages globally [3], is limited, with unique challenges such as Romanized Bangla, code-mixing, code-switching, and the absence of standardized lexicons and corpora. In this paper, we present an innovative approach to perform sentiment and emotion analysis on Bangla comments from YouTube video gaming channels. YouTube video gaming channels, as popular platforms for entertainment and information among Bangla speakers, offer a dynamic space for expressing opinions, feelings, and reactions through comments [4].Our focus shifts to the vibrant world of YouTube gaming channels, a beloved platform for Bangla speakers to engage with content, share opinions, and express emotions through comments. However, analyzing sentiments and emotions in Bangla comments presents its own set of challenges. The unique linguistic characteristics of Bangla, including Romanization and code-switching, pose challenges in sentiment and emotion analysis [1]. We address these challenges through a comprehensive dataset comprising 98,000 Bangla comments from 50 diverse YouTube video gaming channels, covering various game genres [1]. Annotation includes three sentiment levels (positive, negative, neutral) [2]. Our proposed deep learning model is designed to leverage both textual and visual features for effective sentiment and emotion classification. Evaluation metrics encompass accuracy, F1-score, precision, and recall, with comparisons against baselines and state-of-the-art methods. A statistical analysis using the Friedman test gauges results in significance. The following are the contributions of the proposed work:

1) A dataset of 45,722 Bangla comments from YouTube video gaming channels, annotated for sentiment and emotion.
2) Proposed a deep learning-based approach (stacking model) to build a multilabel sentiment analyzer with 96% accuracy.
3) A comprehensive evaluation with various metrics, baselines, and explainable AI techniques

The paper unfolds as follows: Section II reviews related works. Section III outlines our methodology. Section IV presents experimental results and analysis. Section V concludes and explores future directions.

II. RELATED WORKS

Sentiment analysis or opinion mining has become a major point of focus in natural language processing, with over 7,000 articles written on the subject. In recent period, deep learning applications have exhibited impressive performance across different NLP tasks.

[1] Nafis Irtiza Tripto and Mohammed Eunus Ali research on “Detecting Multilabel Sentiment and Emotions from Bangla YouTube Comments”. On their research they use different learning methods and get 49% to 59% accuracy on emotion analysis. Their model can detect six major emotion categories: anger, disgust, fear, joy,

sadness and surprise

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[2] Another work done by Nitish Ranjan Bhowmik, Mohammad Arifuzzaman, M. Rubaiyat Hossain Mondal1, M. S. Islam on “Bangla Text Sentiment Analysis Using Supervised Machine Learning with Extended Lexicon Dictionary” they proposed a specific domain-based categorical weighted lexicon data dictionary (LDD) is developed for analyzing sentiments in Bangla language.

[7] Tan, Jie Ying, Andy Sai Kit Chow, and Chi Wee Tan. ”Sentiment Analysis on Game Reviews: A Comparative Study of Machine Learning Approaches.”They explores sentiment analysis on game reviews through a comparative study of different machine learning approaches. By examining various methods, the research likely provides insights into the effectiveness of different techniques in analyzing sentiments expressed in game reviews. This comparative analysis could contribute to advancements in sentiment analysis methodologies and potentially offer valuable insights for industries reliant on understanding customer feedback, such as the gaming industry.

[8] Ardianto, Rian, et al. ”Sentiment analysis on E-sports for education curriculum using naive Bayes and support vector machine.”The model mentioned in the title appears to be focused on sentiment analysis in the context of e- sports for educational curriculum. It utilizes two machine learning algorithms: Naive Bayes and Support Vector Machine (SVM). These algorithms are commonly used for text classification tasks like sentiment analysis. The study likely aims to analyze sentiments expressed in e-sports-related content within educational contexts, potentially to understand attitudes, perceptions, or impacts on learning outcomes.

[9] Silva, Paula Alexandra, and Renato Santos. ”An opinion mining methodology to analyse games for health.” They introduces an opinion mining methodology designed to analyze games for health purposes. This methodology likely involves using natural language processing and sentiment analysis techniques to extract and analyze opinions expressed in relation to health-focused games. By applying opinion mining techniques, the study aims to provide insights into the effectiveness, user perceptions, and potential impact of games designed for health-related purposes. This research could contribute to the development and evaluation of gamified interventions for health promotion, rehabilitation, or therapeutic purposes.

[10] Gajanayake, G. M. H. C., and T. C. Sandanayake. ”Trending pattern identification of youtube gaming channels using sentiment analysis.”The research focuses on identifying trending patterns of YouTube gaming channels using sentiment analysis. By analyzing sentiment expressed in comments, likes, dislikes, and other engagement metrics, the study likely aims to uncover patterns indicative of popular gaming content on YouTube. This analysis could offer valuable insights into viewer preferences, trends in gaming content consumption, and the factors influencing the success of gaming channels on the platform. Understanding these patterns could be beneficial for content creators, marketers, and platform administrators seeking to optimize their strategies for engaging with gaming audiences on YouTube.

III. PROPOSED METHODOLOGY

Our main objective is to understand user sentiment in Bangla comments on YouTube gaming videos. To achieve this, we will employ sentiment analysis techniques to analyze the sentiment of Bangla comments and gain insights into how viewers perceive and react to different gaming content in the Bangla-speaking community. Additionally, we aim to identify common themes or patterns in positive and negative comments to provide feedback for content creators on how toimprove their videos or better engage their audience. We have utilized an appropriate dataset, the details of which are summarized in Fig 1. This dataset will serve as the foundation for implementing our proposed method, enabling us to conduct sentiment analysis and extract meaningful insights.

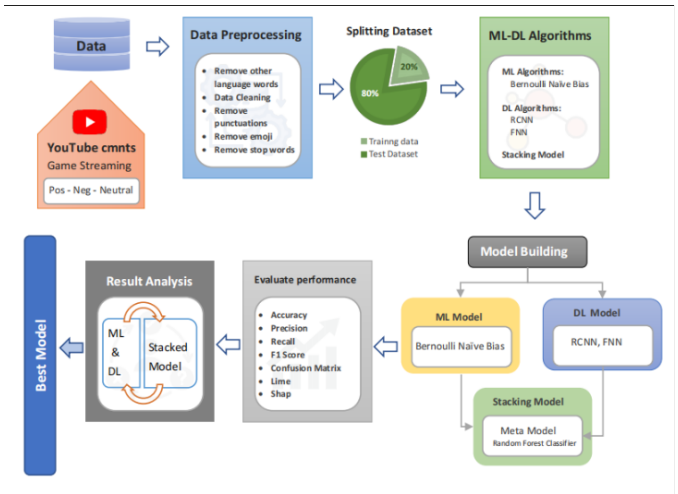

Figure 1. Sequential steps for the development of the proposed model.

A. Data Collection

For our study, we delved into the comments section of over 50 Bangla YouTube gaming review channels. We aimed to capture the diverse conversations and opinions from the Bengali gaming community. Channels like গেমিং উইথ জিহাদ (Gaming with Jihad), তালহা গেমিং (Talha Gaming), and মি.ত্রিপল আর (Mr. Triple R) etc. were among those we explored.In our quest, we gathered a total of 45,722 comments. We were selective, ensuring that only comments written in Bengali were included, maintaining

Table I
SUMMARIZED RELATED WORK

Name	Year	Category	Used dataset	Applied algorithm	Evolution	Advantage	Disadvantage
Salim Sazzad[11]	2019	Gaming review	2489 Comment	LR RR SVM RF ET LSTM	70.9% 69.6% 70.1% 69.1% 68.3% 59%	Performance Improvement	Potential Loss of Nuance
G.M.H.C. Gajanayake[10]	2020	Identification of YouTube gaming channel	500 videos	LR SVM RF NB	71.9% 80% 79% 80.4%	1.Clear Objective 2.Methodological Approach	Lack of detailed analysis
Rian Ardianto[8]	2020	E-sports for education curriculum	15,000 Comment	NB SVM	70.32% 66.92%	1.Clear Presentation of Results 2. Identifying Areas for Future Research	1. Limited Analysis of Results 2. Limited Generalizability
Jie-Ying Tan[7]	2021	Game Review	17543 Comment	LR MNB SVC XGB MLP	79.30% 67.4% 87.7% 79.70% 86.8%	Improved Model Performance	1.Lack of External Validation 2.Assumption of Causality
Paula Alexandra Silva[9]	2022	Analise game for health	538,700 Comment	Usability UX H-QOL	60% 86% 75%	Innovative Approach	1.Limitations in Sarcasm Recognition 2.Focus-on Specific Game

the authenticity and relevance of our dataset. Below is a summarized overview of the collected comments dataset:

Table II
DATA DISTRIBUTION

Sentiment	Number of Comments	Total Comments
Positive	16,839	45,722
Negative	16,169	
Neutral	12,710	

After collecting this vast dataset, we distributed them into positive, negative, and neutral categories. This categorization helped us understand the sentiment balance within the comments and provided a structured framework for our analysis.

B. Data Compilation

Having gained a clear understanding of the dataset's structure and content from Table 2, our next step was data compilation. To facilitate effective model training and evaluation, we partitioned our dataset into two distinct subsets: a training set and a testing set. Utilizing the widely adopted holdout method, we allocated 80% of the data, approximately 36,577 comments, for training the model. This sizable training set was instrumental in enabling our machine learning model to learn from a diverse range of comments, ensuring robustness and generalization capability. The remaining 20% of the dataset, equivalent to 9,144 comments, was reserved for testing the model's performance. This testing set served as an independent benchmark to evaluate the model's accuracy, precision, recall, and other performance metrics. By maintaining this separation between the training and testing sets, we aimed to mitigate the risk of overfitting and ensure that our model's performance could be reliably assessed on unseen data.To further refine our data and prep it for analysis, we took a few steps. We cleaned up the text to get rid of any unwanted characters or symbols. Next, we removed stop words to focus on the meaningful words. We also tokenized the text to break it down into individual tokens. These preprocessing steps helped us get the data in a shape that our sentiment analysis model could work with effectively.

C. Data Preprocessing

Data preparation involves cleaning and formatting the data to ensure its accuracy, consistency, and usability for good performance. Our collected dataset contains some unwanted things like spelling mistakes, punctuation errors, emoji, special characters, and so on. To clean our dataset, we have applied several approaches such as tokenization, non-Bengali word removal, emoji, special character removal, cleaning, URL removal, etc.

1) Removing Non-Bengali Words, Special Characters, Emoji, URLs, and Punctuation: After collecting data from the social platform, we also have irrelevant things such as Non-Bengali Words, Special Characters, Emoji, URLs, and Punctuation, and so on in our dataset. So, we cleaned our dataset by removing those things so that the dataset can perform smoothly.

2) stop Word Removing: Stop word removal is the process of eliminating commonly used words, known as "stop words". Every language has stop word as Bengali. In Bengali language (Etc), (And), (This) and so on are known as Bengali Stop words. Because these words are frequently used in the Bengali language.

3) Tokenization: Tokenization refers to the process of converting a sequence of text into smaller parts, known as tokens. By applying tokenization, the dataset is split into small parts so that spelling mistakes, punctuation errors, emojis, special characters, and so on can be detected.

4) Lowercasing: Convert all text to lowercase. This step ensures consistency in the dataset, as "Hello" and "hello" should be treated as the same word.Lowercasing helps avoid duplication of words due to different capitalization.

5) Handling Missing Values: Address any missing data (NaNs) in your dataset. Common approaches include imputation (replacing missing values with estimated values) or removing rows with missing data.

6) Normalization: Normalize numerical features to a common scale. Techniques like min-max scaling or z-score normalization can be useful. Normalization ensures that features with different scales contribute equally during analysis.

7) Removing Duplicates: Check for duplicate records in your dataset. Duplicate data can skew results and lead to overfitting

8) Handling Noisy Text: Sometimes text data contains noise, such as typos or inconsistent spellings. Consider using spell-checking tools or custom rules to clean up noisy text.

9) Feature Engineering: Create new features from existing ones if relevant. For example, extracting features like word count, character count, or sentiment scores can enhance model performance.

D. Classification Model

We have developed and trained a Bangla document classification model specifically tailored for analyzing YouTube gaming videos. Our study utilized the Multinomial Naive Bayes (NBN), Recurrent Convolutional Neural Network (RCNN), and Feedforward Neural Network (FNN) models to classify and analyze the content of these gaming videos. Following training, we rigorously evaluated the performance of our models using various metrics on a dedicated test dataset. The Bernoulli Naive Bayes (BNB) model is another powerful classifier commonly used in text-based tasks, such as analyzing YouTube gaming video descriptions or titles [12]. Similar to the Multinomial Naive Bayes model, BNB is straightforward and efficient, making it suitable for scenarios with limited computational resources. In contrast, the Recurrent Convolutional Neural Network (RCNN) combines the spatial understanding of Convolutional Neural Networks (CNNs) with the sequential processing capabilities of Recurrent Neural Networks (RNNs). This fusion allows RCNN to capture both local and global context from sequential data, such as video transcripts or comments, making it ideal for understanding the narrative flow within gaming videos. Meanwhile, the Feedforward Neural Network (FNN), also known as a multilayer perceptron (MLP), offers versatility and adaptability, making it suitable for various machine-learning tasks. In the context of YouTube gaming video analysis, FNNs excel in handling structured data and deciphering complex, nonlinear relationships within the video content. Each model contributes unique strengths, catering to different aspects of analyzing Bengali YouTube gaming videos efficiently and effectively.

IV. Result Analysis and Discuss

The primary emphasis of the proposed study revolves around accurately discerning sentiments in Bangla game reviews through the utilization of learning models. Additionally, it delves into elucidating the rationale behind the varying performance outcomes, employing explainable NLP techniques. The paper further elaborates on the performance metrics, result analysis, Friedman test statistics, and the application of explainable NLP. The provided classification report presents the performance of a Bernoulli Naive Bayes (BNB) machine learning model on a dataset with three classes: Negative, Neutral, and Positive. The model achieves an overall accuracy of 60%, with notable precision for Negative (0.55) and Neutral (0.68) classes. However, it exhibits higher recall for Negative (0.79) compared to Neutral (0.43) and Positive (0.55) classes. The F1-scores reflect a balanced performance, with Negative class having the highest score (0.65). Although the model's accuracy is reasonable, there is room for improvement, especially in accurately predicting instances of Neutral and Positive classes. However, it exhibits higher recall for Negative (0.79) compared to Neutral (0.43) and Positive (0.55) classes. The F1-scores reflect a balanced performance, with Negative class having the highest score (0.65). Although the model's accuracy is reasonable, there is room for improvement, especially in accurately predicting instances of Neutral and Positive classes.

Table III
BNB Model Result Summery

Category	Precision	recall	f1-score	support
Negative	0.55	0.79	0.65	3191
Neutral	0.68	0.43	0.53	2563
Positive	0.63	0.55	0.59	3390
Accuracy			0.60	9144
Macro Average	0.62	0.59	0.59	9144
Weighted Average	0.62	0.60	0.59	9144

The provided classification report outlines the performance metrics of a BNB (Bernoulli Naive Bayes) model applied to a dataset with three distinct categories: Negative, Neutral, and Positive. Despite achieving an accuracy of 60%, the model's precision, recall, and F1-scores vary across the classes. In particular, the Negative class exhibits a precision of 0.55, recall of 0.79, and an F1-score of 0.65. For the Neutral class, the precision is 0.68, recall is 0.43, and F1-score is 0.53. Lastly, the Positive class displays a precision of 0.63, recall of 0.55, and an F1-score of 0.59. Both the macro and weighted averages for precision, recall, and F1-score are lower compared to the RCNN model, standing at 0.62, 0.59, and 0.59, respectively. These results suggest that while the BNB model demonstrates some capability in classification, it may struggle with accurately capturing the nuances present in the dataset, leading to comparatively lower performance metrics.

Table IV
RCNN Model Result Summery

Category	Precision	recall	f1-score	support
Negative	0.93	0.92	0.93	3191
Neutral	0.85	0.82	0.83	2563
Positive	0.85	0.88	0.87	3390
Accuracy			0.88	9144
Macro Average	0.88	0.87	0.88	9144
Weighted Average	0.88	0.88	0.88	9144

The provided classification report summarizes the performance of an RCNN (Recurrent Convolutional Neural Network) model on a dataset with three classes: Negative, Neutral, and Positive. The model achieves an impressive accuracy of 88%, with high precision, recall, and F1-scores across all classes. Specifically, Negative class has precision, recall, and F1-score of 0.93, 0.92, and 0.93 respectively; Neutral class scores 0.85, 0.82, and 0.83; and Positive class scores 0.85, 0.88, and 0.87 respectively. The macro and weighted averages for precision, recall, and F1-score are also high at 0.88, reflecting the model's strong overall performance. These results suggest that the RCNN model effectively captures the nuances of the dataset and accurately predicts class labels.

Table-5: FNN Model Result Summery

The provided classification report outlines the performance metrics of an FNN (Feedforward Neural Network)

Category	Precision	recall	f1-score	support
Negative	0.91	0.92	0.91	3191
Neutral	0.86	0.78	0.82	2563
Positive	0.84	0.89	0.86	3390
Accuracy			0.87	9144
Macro Average	0.87	0.86	0.86	9144
Weighted Average	0.87	0.87	0.87	9144

model on a dataset with three classes: Negative, Neutral, and Positive. The model demonstrates an overall accuracy of 87%, with solid precision, recall, and F1-scores across all classes. Specifically, Negative class achieves precision, recall, and F1-score of 0.91, 0.92, and 0.91 respectively. Neutral class scores 0.86, 0.78, and 0.82, while Positive class achieves 0.84, 0.89, and 0.86 respectively. Both macro and weighted averages for precision, recall, and F1-score are consistent at 0.87, indicating a balanced performance across all classes. These results suggest that the FNN model effectively learns patterns within the dataset and accurately predicts class labels, albeit with slightly lower performance compared to more complex architectures like RCNN.

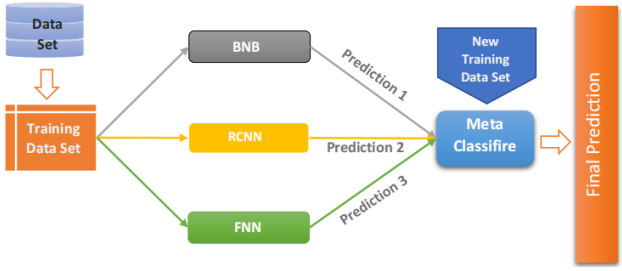

Figure 2. Stacking Model

Table V
Stacking Model summery

Category	Precision	recall	f1-score	support
Negative	0.98	0.98	0.98	3191
Neutral	0.96	0.93	0.95	2563
Positive	0.95	0.97	0.96	3390
Accuracy			0.96	9144
Macro Average	0.96	0.96	0.96	9144
Weighted Average	0.96	0.96	0.96	9144

The provided classification report presents the performance metrics of a hybrid model, specifically a Stacked Model, which combines the predictions from Bernoulli Naive Bayes (BNB), Recurrent Convolutional Neural Network (RCNN), and Feedforward Neural Network (FNN) models. This stacked model achieves an impressive accuracy of 96%, showcasing its effectiveness in predicting class labels. Across all classes, the precision, recall, and F1-scores are consistently high. For Negative class, precision, recall, and F1-score are 0.98, 0.98, and 0.98 respectively. Neutral class achieves 0.96, 0.93, and 0.95, while Positive class scores 0.95, 0.97, and 0.96 respectively. Both macro and weighted averages for precision, recall,

Table VI
COMPARISON AMONG MODELS

Model	Accuracy	AUC
BNB	60%	0.75
RCNN	88%	0.81
FNN	87%	0.84
Stacking	96%	0.97

and F1-score stand at 0.96, underscoring the balanced and robust performance of the Stacked Model. These results indicate that the combination of diverse modeling techniques has effectively captured the complexities of the dataset, resulting in highly accurate predictions across all classes.

The comparison table showcases the performance of four different models: Bernoulli Naive Bayes (BNB), Recurrent Convolutional Neural Network (RCNN), Feedforward Neural Network (FNN), and a Stacked Model combining BNB, RCNN, and FNN. The Stacked Model outperforms the individual models with an accuracy of 96%, demonstrating superior precision, recall, and F1-scores across all classes. RCNN follows closely with an accuracy of 88%, while FNN achieves an accuracy of 87%. BNB trails behind with an accuracy of 60%. The Stacked Model's exceptional performance highlights the efficacy of combining diverse modeling techniques for improved classification accuracy.

A. Explainable AI

1) Lime: The provided diagram illustrates sentiment analysis related to YouTube gaming. On the left, a bar graph labeled "Prediction probabilities",displays prediction probabilities for different sentiments:

- Negative: Very Low (0.01)
- Neutral: Moderate (0.23)
- Positive: High (0.76)

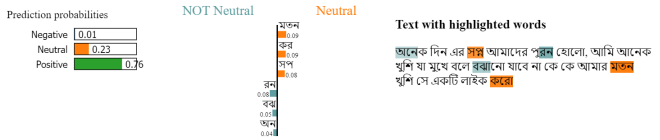

Figure 3. Lime explanation for the model

These probabilities represent the sentiment score assigned to the analyzed text.In the middle section, two columns labeled "NOT Neutral" and "Neutral" likely represent specific features associated with sentiment values(ranging from 0.04 to 0.09).On the right side, there's a section labeled "Text with highlighted words."A detailed analysis of the highlighted words is, "অনেক,স□,রন,বঝা,মতন,কেরা।" Overall, the diagram helps analyze sentiment.

2) : SHAP: The diagram represents predictions made by three different machine learning models: Bernoulli Naive Bayes (BNB), Recurrent

Convolutional Neural Network (RCNN), and Feedforward Neural Network (FNN).Each of these models tries to guess something related to YouTube games. The x-axis (the horizontal line) represents something called "SHAP interaction value." Think of it as a scale from -0.5 to 0.5. This value helps us understand how much different factors influence the models' predictions.In this diagram we show that, the blue and red crosses on each plot represent the predictions made by the models.The middle point of each cross shows the average guess.The lines extending horizontally from the crosses indicate the range or spread of predictions.So,if the lines are short, it means the models were pretty confident in their guesses. If they're long, they were less sure.

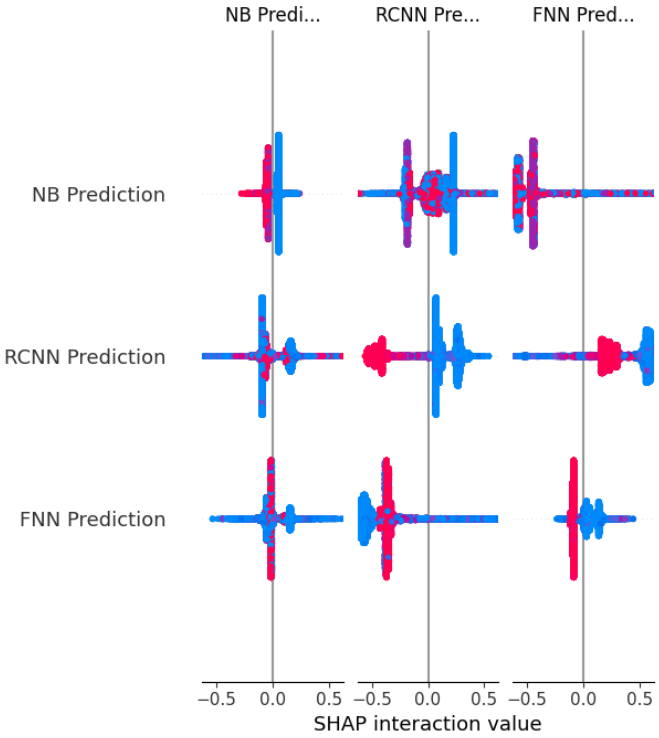

Figure 4. SHAP explanation for the model

V. CONCLUSION AND FUTURE WORK

Analyzing sentiments in Bengali online video gaming is crucial, but there's a lack of benchmark datasets and research in this area. In our study, we created a dataset of 45,722 Bengali online gaming reviews from various You Tube Gamming channels. We manually annotated them, focusing on accurately identifying sentiments using learning models. We employed SMOTE to enhance model performance, with ML and DL models showing average improvements of 1% and 2% respectively. RCNN and

Stacking models outperformed others, achieving the highest accuracies of 88% and 96%. Hybrid deep learning methods surpassed base methods.Future plans include enriching and balancing the dataset, exploring hybrid feature extraction techniques, implementing transformer-based learning, and trying different hybrid methods.

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